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INTERNAL BOUNDARY LAYERS: I. HEIGHT FORMULAE FOR NEUTRAL AND DIABATIC FLOWS

SERGIY A. SAVELYEV* and PETER A. TAYLOR
Centre for Research in Earth and Space Science, York University, Toronto, Ontario, Canada, M3J 1P3

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Abstract. The notion of an internal boundary layer (IBL) appeared in studies of local advection within the atmospheric boundary layer when air flows over a change in surface conditions. These include surface roughness, thermal and moisture properties. An ability to predict the height of the IBL interface in the atmosphere under neutral stability, accompanied by certain assumptions on the form of the mean flow parameters, have been a means of obtaining information on the velocity profile after step changes in roughness for more than half a century. A compendium of IBL formulae is presented. The approach based on the 'diffusion analogy' of Miyake receives close attention. The empirical expression of Savelyev and Taylor (2001, Boundary Layer Meteorol. 101, 293–301) suggested that turbulent diffusion is not the only factor that influences IBL growth. An argument is offered that an additional element, mean vertical velocity or streamline displacement, should be taken into account. Vertical velocity is parameterized in terms of horizontal velocity differences employing continuity constraints and scaling. Published data are analyzed from a new point of view, which produces two new neutral stratification formulae. The first implies that the roughness lengths of adjacent surfaces are equally important and a combined length scale can be constructed. In addition new formulae to predict the height of the region of diabatic flow affected by a step change in surface conditions are obtained as an extension of the neutral flow case.

Keywords: Diffusion analogue, Internal boundary layer, Surface heterogeneity.

1. Introduction

The breakthrough concept of a boundary layer in a fluid flow adjacent to a surface was introduced by Prandtl at the beginning of the twentieth century and facilitated the development of scientific tools for practical studies. The atmospheric boundary layer (ABL) can be defined by the extent of diurnal influence in the air caused by the underlying surface. This is different from Prandtl's concept, but still has the property that horizontal scales are much greater than vertical scales close to the boundary. The concept of an internal boundary layer (IBL) involves a process of adjustment to new surface conditions (with a transition from the old state to a new one) and the word

* E-mail: savelyev@yorku.ca

'internal' reflects the fact that this new boundary layer develops within the existing ABL. Garratt (1990, 1992) provided a good review of IBL treatment.

As one example, internal boundary layers are formed in coastal regions when air flows from sea to land, and vice versa, and are a notable feature of a coastal climate. They influence physical processes on a variety of scales starting from microscale air–sea exchange up to interaction with sea–land breezes and synoptic–scale fronts passing through the domain. Land–water (water–land) transition is characterized by a change in surface roughness that is usually accompanied by temperature and flux differences, and that in turn leads to modification of atmospheric stability.

There are many practical applications involving, for example, air quality, wind energy and marine ecology. It was recognized long ago that, in order to be representative of general conditions rather then a local microclimate, a weather station should be placed at a certain distance from obstacles and changes in surface cover to avoid the so-called 'leading-edge effect', i.e. it should reside within an internal boundary layer in a region where flow is in a state close to the new equilibrium. The 'leading-edge' is related to a line of relatively sharp changes in surface conditions. The estimation of fluxes (of momentum, heat, water vapour) within these transition zones is also important, for example, in the parameterization of heterogeneous domains in larger scale models. Simple models based on the internal boundary-layer concept are in extensive use by the wind engineering community (see, for example, WMO, 1981; Barthelmie and Palutikcof, 1996; Castino and Tombrou, 1998; or the simple guidelines of Walmsley et al., 1989). Research efforts aimed towards further development of the 'Simple Guidelines' model led to results presented in this article for a computationally inexpensive model of predicting the IBL height within both the neutral and diabatic near-surface atmospheric boundary layer.

2. The Internal Boundary Layer – A Review of the Structure

Horizontal homogeneity and steady state assumptions are useful concepts for understanding processes in the constant-flux surface layer. In real life this situation is of rare occurrence. Forcing in the atmospheric boundary layer varies on an abundance of time and space scales. The surface underlying the atmospheric flow is more often than not inhomogeneous, changing continuously and often abruptly. These changes often involve a change in surface temperature and/or humidity and in the vertical fluxes, as for example in the case of a land-to-water transition. The mean temperature and water vapour profiles are affected as well as the mean wind speed profile and the turbulence. Horizontal gradients that appear as a consequence cause a transport of flow variables referred to as 'local advection' (Garratt, 1992). The part of the

atmosphere adjacent to the surface where the influence of the new surface conditions is detected is termed the internal boundary layer.

In many cases solutions to the IBL problem are put into the context of the simplest possible case when two distinct semi-infinite horizontally homogeneous surfaces border each other along a straight line. This context renders the problem two-dimensional and allows us to make use of knowledge about turbulence structure and mean flow parameters in the constant-flux surface layer. Figure 1 helps to visualize the simplified picture of a two-dimensional (2D) internal boundary layer, of depth $\delta(x)$. Note that throughout the article we use subscripts U and D to denote parameters that belong to upstream and downstream surfaces, respectively. For instance, z_{0U} is used for the roughness length of the upstream surface while z_{0D} is used for the downstream surface roughness length.

One may expect that, if the fetch above the new surface is long and conditions are constant, the flow would achieve a new equilibrium. The question arises: how exactly does this happen? Many scientists believe that the equilibrium layer first appears close to the surface, grows in thickness with fetch and eventually becomes a new constant-stress surface layer. Taylor (1969) however noted that full adjustment, including wind direction, involves the entire ABL and requires a fetch of order U/f where f is the Coriolis parameter. For $f = 10^{-4}$ s^{-1} and $U = 10$ m s^{-1} this distance is 100 km. Ignoring these subtle points we note that the equilibrium layer was reported to exist in various laboratory experiments and numerical simulations. Hence we can describe a general structure of the flow after the change in surface conditions as consisting of an equilibrium layer immediately above the surface, a transition layer and the outer region of incoming flow formed above the old surface. The internal boundary layer comprises the first two layers. It is, however, not obvious how to confirm the emerging equilibrium state.

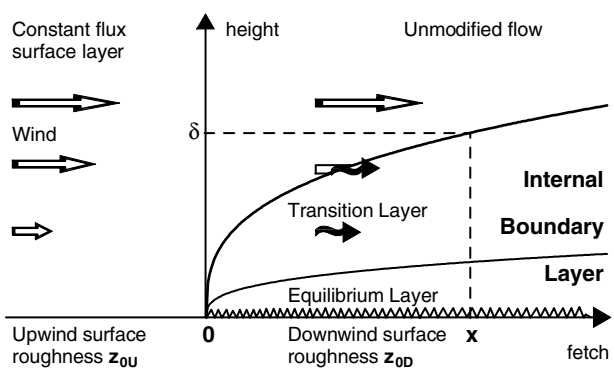

Figure 1. Sketch of a two-dimensional internal boundary layer developing within a constant-flux surface layer after a change in surface conditions. δ is the height of the interface at distance x from a leading edge.

It should be noted that there is a certain degree of ambiguity in the literature concerning the term 'internal boundary layer'. Some scientists relate this term to the newly formed boundary layer that has already achieved a certain degree of equilibrium with the new surface, i.e., to the 'equilibrium' layer of the above description. The transition zone is considered to be a separate layer between two equilibrium ones. Such a definition was adopted, for example, in the model of Deaves and Harris that was the core of the United Kingdom design guidance and wind load code (Cook, 1997). We will however, stay with the definition given first and distinguish between the IBL and equilibrium layer heights as in Figure 1.

The definition of the IBL comprises the idea of modification of the flow characteristics due to the impact of changing conditions at the lower boundary. Any flow parameter that can be tracked from the upwind region into the downwind one will serve as an indicator of the interface between the undisturbed and modified flow. In field studies the spatial distribution of some characteristic or primitive variable is costly to obtain. Measurements of vertical profiles at a limited number of downwind locations are in most cases all that are available. It appears that the outward propagation speed of a particular field modification is different for various fluid characteristics. Even if we restrict ourselves to the parameters that are governed by the same physical processes, e.g., mean wind speed and shear stress, one can obtain 'velocity boundary-layer' and 'stress boundary-layer' heights that differ by a factor of almost two (Shir, 1972).

Mean wind speed profiles are probably the easiest measurements to obtain and the most common way to define the IBL. By comparing two profiles that belong to upwind and downwind regions one can deduce the IBL height by finding the height where two wind speed values differ by not more than a specified amount. The reference upwind profile is usually obtained right at the point where surface conditions change or close to it, but the question arises whether this reference profile has been modified from the upstream form already. In some cases a wind speed profile is available at one point only. If it spans the entire IBL and part of the unmodified flow such a profile can be used in the analysis. To obtain the 'traditional' IBL height one can look for intersection of the lower part of the profile, which may have a new surface log–law equilibrium form, and the top portion reflecting the log law of the old surface (left panels in Figure 2). There is a caveat that if no reference measurements are made in the upwind region then streamline displacement and IBL transition effects cannot be clearly separated. Alternately, in the case of a single profile, one could extend the top logarithmic part into the transition region down to the inflection point (see right panels in Figure 2) and obtain lower values for the IBL height, denoted here by $\delta_{1/2}$. It has to be mentioned that some researchers place the IBL top somewhere in the transition region (Logan and Fichtl, 1975 called it a buffer layer) between

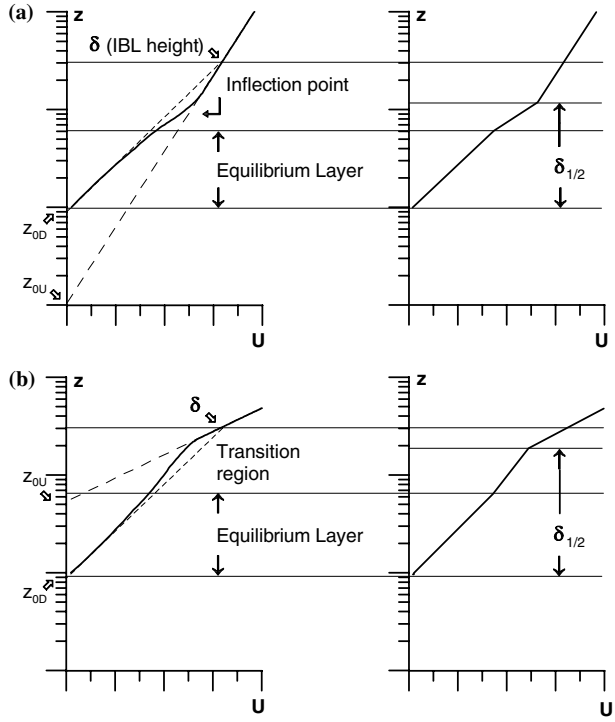

Figure 2. Sketches of the wind speed profile (left panel) at some distance from a leading edge and possible definitions of the IBL height near the inflection point of the profile (right panel); smooth-to-rough (a) and rough-to-smooth (b) transitions.

a well discernible internal equilibrium layer and an upper layer influenced by the upwind surface only, i.e., lower than according to the conventional definition. Logan and Fichtl (1975) stated that the IBL interface separates two regions. For the rough-to-smooth transition, one starts from the ground and comprises the accelerating fluid while the other extends up to unmodified flow and comprises decelerating fluid. This definition was adopted by Jegede and Foken (1999) while analyzing data from field campaigns. Their empirical formula $\delta = 0.09x^{0.8}$ (see Table I for details of many of the IBL formulae discussed herein), which fits data derived from the mean wind speed profiles in a range of fetches from 140 to 260 m, produces an IBL height that is considerably lower than that implied by other formulae (see Figure 3). Elliott's (1958) model with a dependance on $M = \ln(z_{0D}/z_{0U})$ predicts the highest IBL in both rough-to-smooth ($M = -1.3$) and smooth-to-rough ($M = +1.4$) cases, closely followed by Miyake's (1965) approach. Jackson's (1976) formulation differs from Miyake's, not only by the value of the constant and roughness scale, but also by the definition of the modified region height. Expressions of Wood (1982) and Pendergrass and Aria (1984)

TABLE I

Short fetch IBL height formulae (in chronological order). IBL height δ and distance from a leading edge x are in metres.

Formula	Author(s)	Notes
$\frac{\delta}{z_{0D}} = (0.75 - 0.03M)\left(\frac{x}{z_{0D}}\right)^{0.8}$	Elliott (1958)	$M = \ln\frac{z_{0D}}{z_{0U}}$
$4\kappa^2(\frac{x}{z_{0D}} - \frac{x_0}{z_{0D}}) = \frac{\delta}{z_{0D}}\left[\delta\ln\frac{\delta}{z_{0D}} - 5 + \frac{M}{2} + \frac{4-7M/6-M^2/4}{\ln(\delta/z_{0D})-1+M/4} + \frac{4+7M/6+M^2/24+M^3/16}{(\ln(\delta/z_{0D})-1+M/4))^2}\right]$, $x_0 = 0$ for $-M \le 3$ else $4\kappa^2\frac{x_0}{z_{0D}}e^{M-1} = -\frac{64}{27} - \frac{M}{18} - \frac{88}{27M} + \frac{64}{9M^2}$	Panofsky and Townsend (1964)	z_0 is the surface roughness length, κ is the von Karman const.
$2\kappa^2 x = \delta\ln\frac{\delta}{z_{0D}}$	Townsend (1965)	
$1.73\kappa\frac{x}{z_{0D}} = \frac{\delta}{z_{0D}}(\ln\frac{\delta}{z_{0D}} - 1) - \frac{\delta_0}{z_{0D}}(\ln\frac{\delta_0}{z_{0D}} - 1)$	Miyake (1965)	$\delta_0 = \delta(x = 0)$
$\kappa^2 x = \delta\frac{\ln\frac{\delta}{z_{0D}}(\ln\frac{\delta}{z_{0D}}+M)}{2\ln\frac{\delta}{z_{0D}}+M}$	Townsend (1966)	$\kappa^2 x = \frac{\delta\ln\frac{\delta}{z_{0U}}\ln\frac{\delta}{z_{0D}}}{2\ln\frac{\delta}{\sqrt{z_{0U}z_{0D}}}}$
$\frac{d\delta}{dx} = \kappa^2\frac{\delta}{z_{0U}}\frac{\ln^2\frac{\delta}{z_{0D}}+\ln\frac{\delta}{z_{0D}}\ln\frac{\delta}{z_{0U}}}{F(\delta)\ln\frac{\delta}{z_{0U}}}$	Radikevitsch (1971)	$\delta(x = 0) = z_{0D}$
$F(\delta) = \frac{\delta}{z_{0U}}\ln^2\frac{\delta}{z_{0D}} - \ln\frac{\delta}{z_{0D}}\left(3\frac{\delta}{z_{0U}} + \frac{z_{0D}}{z_{0U}}\right) + 4\left(\frac{\delta}{z_{0U}} - \frac{z_{0D}}{z_{0U}}\right)$		
$\delta = f_1\left(\frac{z_{0U}}{z_{0D}}\right)x^{(0.8+f_2(z_{0U}/z_{0D}))}$	Shir (1972)	f_1, f_2 are not given
$1.5\kappa\frac{x}{z_{0r}} = \frac{\delta}{z_{0r}}\left(\ln\frac{\delta}{z_{0r}} - 1\right)$	Panofsky (1973)	$z_{0r} = \max(z_{0U}, z_{0D})$
$\frac{\delta}{z_{0D}} = 0.095\left(\frac{x}{z_{0D}}\right)^{1.03}$	Schofield (1975)	Wind-tunnel data
$0.75\kappa\frac{x}{z_0'} = \frac{\delta_0'}{z_0'}\left(\ln\frac{\delta_0'}{z_0'} - 1\right) - \frac{\delta_0'}{z_0'}\left(\ln\frac{\delta_0'}{z_0'} - 1\right)$	Jackson (1976)	$\delta' = \delta - d_D$ d_D – displacement $z_0' = \sqrt{\frac{z_{0U}^2+z_{0D}^2}{2}}$
$\frac{\delta}{z_{0U}} = f_1(M)\left(\frac{x}{z_{0D}}\right)^{0.8}$	Andreopoulos and Wood (1982)	$f_1(M)$ was tabulated elsewhere
$\frac{\delta}{z_{0r}} = 0.28\left(\frac{x}{z_{0r}}\right)^{0.8}$	Wood (1982)	$z_{0r} = \max(z_{0U}, z_{0D})$
$\kappa\frac{x}{z_{0U}} = \frac{\delta}{z_{0D}}\left(\ln\frac{\delta}{z_{0D}} - \ln\frac{z_{0D}}{z_{0U}} - 1\right) + \frac{z_{0D}}{z_{0U}}$	Raabe (1983)	
$\frac{\delta}{z_{0D}} = 0.32\left(\frac{x}{z_{0D}}\right)^{0.8}$	Pendergrass and Aria (1984)	
$1.25\kappa\frac{x}{z_{0D}} = \frac{\delta}{z_{0D}}\left(\ln\frac{\delta}{z_{0D}} - 1\right) + 1$	Panofsky and Dutton (1984)	
$2.25\kappa\frac{x}{z_{0D}} = \frac{\delta^*}{z_{0D}}\left(\ln\frac{\delta^*}{z_{0D}} - 1\right)$	Troen et al. (1987)	
$\delta = C\delta^*$	WASP model	$C = 0.3$ (constant)
$\delta = 0.09x^{0.8}$	Jegede and Foken (1999)	Data fetch range: 140 – 260 m
$\delta\left(\ln\frac{\delta}{z_{0U}} - 1\right) = 1.25\kappa(1 + 0.1M)x$	Savelyev and Taylor (2001)	$M = \ln\frac{z_{0D}}{z_{0U}}$
$\frac{\delta}{z_{0D}} = 10.56\left(\frac{x}{z_{0D}}\right)^{0.33}$	Cheng and Castro (2002)	
$\frac{d\delta}{dx} = C\kappa\left(1 + \frac{\delta}{x}M\right)\left(\ln\frac{\delta}{z_{0U}}\right)^{-1}$	Equation (23)	$C = \sigma_w/u_*$ $(= 1.25)$
$\delta\left(\ln\frac{\delta}{z_{0U}} - 1 - C\kappa M\right) = C_{\kappa x}$	Equation (26)	$2\delta\left(\ln\frac{\delta}{\sqrt{z_{0U}z_{0D}}} - 1\right) = x$

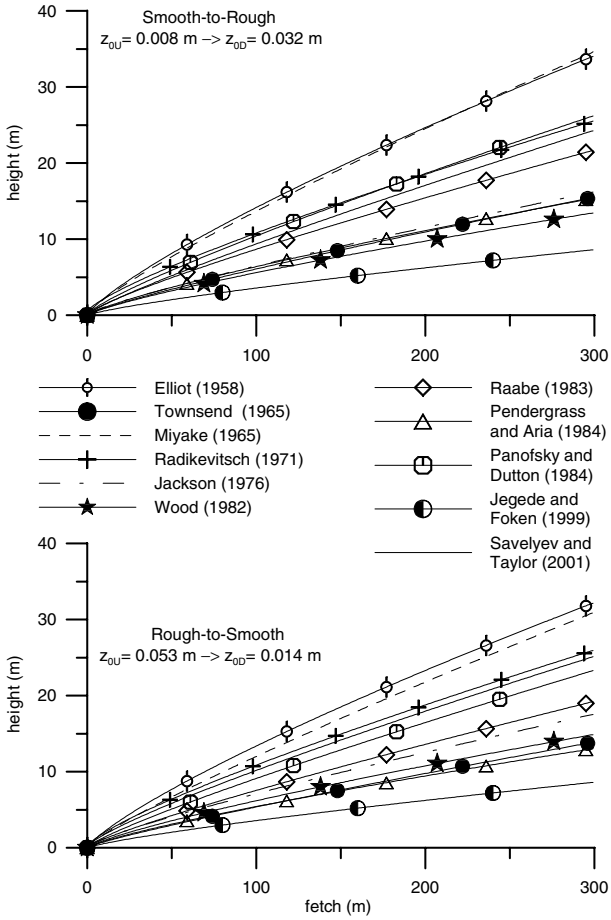

Figure 3. Short fetch IBL height for smooth-to-rough and rough-to-smooth transitions predicted by several formulae. Roughness lengths were selected to match those in the Jegede and Foken (1999) field data. Formulae details are in Table I.

were obtained by fitting to experimentally determined heights, and so are based on the IBL definition pertaining to the experiments. Radikevitsch (1971) offered an approximate analytical solution of a simple 1.5-order model that can be used to forecast the extent of the region of modified adiabatic flow and characteristics of the turbulence inside it. Arguments in Raabe (1983) are somewhat similar to those in 'diffusion analogy' approach: vertical displacement of the air in the IBL is set proportional to a friction velocity. The latter is determined through a slope of assumed wind profile. Later, in Raabe (1991), the same approach was extended to diabatic flow. IBL heights predicted by the various models for 300 m fetch depend on the definition adopted and range from 8 to 34 m.

If there are changes in surface temperature or heat flux, as well as, or instead of, a roughness change, then the notion of an internal boundary layer also applies, at least at short fetch. Potential temperature profiles downwind of the leading edge are expected to consist of portions with distinct slopes that reflect lapse rates before and after the line of separation in surface temperature or heat flux regimes. As in the case of the vertical distribution of wind speed we can look for the intersection of these profile parts on a log-linear plot or for a discontinuity in the profile. A more extensive discussion is available in Savelyev (2003). For a review of the approaches to the prognosis of thermal IBLs, see Garratt (1990, 1992), Melas and Kambezidis (1992) or Källstrand and Smedman (1997).

3. The IBL – A Review of IBL Height Formulae

Perhaps the earliest and most quoted formula used to calculate the IBL height δ is that of Elliott (1958),

$$\frac{\delta}{z_{0D}} = \left(0.75 - 0.03 \ln \frac{z_{0D}}{z_{0U}}\right)\left(\frac{x}{z_{0D}}\right)^{0.8}, \tag{1}$$

where z_{0U} and z_{0D} are the roughness lengths of the upwind and downwind surfaces, respectively. It is a particular realization of the family of formulae that can be expressed in a general form

$$\frac{\delta}{z_1} = f_1\left(\frac{z_{0D}}{z_{0U}}\right)\left(\frac{x}{z_2}\right)^{\alpha}. \tag{2}$$

The form of the function f_1 is supposed to be determined from experiment and in a simple case can just be a constant; the same applies for α, e.g. $\alpha = 0.8 + f_2(z_{0U}/z_{0D})$ in Shir (1972). In most cases α is considered to be close to 0.8. When this equation is used under atmospheric stability conditions other than neutral, diabatic effects are included through the function f_1 and through α. For example, Bergström et al. (1988) proposed

$$\delta = 0.2x^{0.78-0.33z/L}, \tag{3}$$

where L is the Obukhov length and $z = 11$ m. Note that there is no dependence on surface roughness in this case. Here and in other cases values of empirical coefficients shown are for the case when δ and x are in metres. Values for z_1 and z_2 in Equation (2) are usually chosen by considering appropriate scaling. Apart from upwind and downwind roughness lengths z_{0U}, z_{0D} or $z_{0r} = \max(z_{0U}, z_{0D})$, scaling variables equal to $(z_{0U}z_{0D})^{1/2}$ (Deaves, 1981) or $[0.5(z_{0U}^2 + z_{0D}^2)]^{1/2}$ (Jackson, 1976) have been proposed.

Specific examples in the form of Equation (2) are obtained based on scaling considerations, theoretical reasoning or empirical data treatment. For

illustration let us mention formulae from Andreopoulos and Wood (1982) derived on dimensional grounds,

$$\frac{\delta}{z_{0U}} = f_1(M)\left(\frac{x}{z_{0D}}\right)^{0.8}, \tag{4}$$

where $M = \ln(z_{0D}/z_{0U})$. The functional dependance $f_1(M)$ is supposed to be determined empirically and was tabulated elsewhere (see references in the paper cited). Note that Wood (1982) argued that only the rougher surface is important and suggested

$$\frac{\delta}{z_{0r}} = 0.28\left(\frac{x}{z_{0r}}\right)^{0.8}, \tag{5}$$

where $z_{0r} = \max(z_{0U}, z_{0D})$. We list many of the proposed equations to predict the IBL height that can be found in the literature in Table I. It is worth mentioning that similarity considerations limit their validity to a region close to the surface. Thus Wood (1982) set this limit at δ/z_i equal to approximately 0.2, where z_i denotes the thickness of the boundary layer within which the IBL develops. In many other papers the formulae are considered effective only in an atmospheric 'constant-flux' surface layer ($\approx 0.1z_i$). It is obvious that confinement to a wall region restricts the fetch range as well (for long fetches the IBL will grow out from the surface layer). An assumption of relatively small perturbations (e.g., not more than a moderate roughness change) is also often present or implied.

The above mentioned limitations or assumptions hold for another distinct approach to the derivation of the IBL height, namely the 'diffusion analogy' approach. It is attributed to Miyake (1965) (see Brutsaert, 1982 or Jackson, 1976). Miyake first combined Elliott's idea of an analogy between a spread of impact from a roughness change and a spread of a smoke plume with a theory describing the turbulent diffusion in the surface layer by means of similarity concepts due to Monin and Obukhov (1954). The analogy implies that propagation of the influence of a surface change is similar to the spread of a passive contaminant. An excess or deficit of a turbulent quantity that is a result of small changes in surface conditions then diffuses upwards in the same way that a pollutant (or smoke) would. The rate of growth of the affected region or its height in the two-dimensional (2D) case is proportional to a vertical diffusion intensity expressed by $\sigma_w = (\overline{w'w'})^{1/2}$, i.e.

$$\frac{d\delta}{dt} = A\sigma_w. \tag{6}$$

After expanding $d\delta/dt = \partial\delta/\partial t + (\partial\delta/\partial x)(dx/dt)$ we have, for steady state, $d\delta/dt = (\partial\delta/\partial x)(dx/dt)$ and assuming that dx/dt equals the mean wind speed we obtain the equation for the interface as

$$U(\delta)\frac{d\delta}{dx} = A\sigma_w. \tag{7}$$

Miyake applied the flux–variance relationship for the neutrally stratified surface layer, namely $\sigma_w/u_* =$ constant, denoted by C, to obtain

$$U(\delta)\frac{d\delta}{dx} = Bu_*. \tag{8}$$

This can be integrated provided one has a knowledge of $U(z)$ and u_* or their ratio. Miyake's result was

$$1.73\kappa\frac{x}{z_{0D}} = \frac{\delta}{z_{0D}}\left(\ln\frac{\delta}{z_{0D}} - 1\right) - \frac{\delta_0}{z_{0D}}\left(\ln\frac{\delta_0}{z_{0D}} - 1\right), \tag{9}$$

where δ_0 is an IBL height at $x = 0$ and $\kappa = 0.4$ is the von Karman constant. A number of scientists followed this path offering their reasoning for the functional form of the $U(\delta)$, proper scaling and the value of the constant B (see Panofsky, 1973; Jackson, 1976; Panofsky and Dutton, 1984). Their formulae are listed in Table I.

3.1. Observations and wind-tunnel experiments

Observations of the IBL height made during experiments are a source of empirical expressions on the one hand and a means to check theoretical ideas on the other. Many such checks are based primarily on a dataset that encompasses many atmospheric and wind-tunnel experiments of the 1960–1970 period. It was used by Jackson (1976) to test his prediction of the IBL growth with fetch. References to original works are contained therein. Walmsley (1989) augmented this dataset with measurements of Peterson et al. (1979) and Taylor (1969). This inclusion extended the longest recorded fetches to 90–160 m (from around 50 m in the original dataset). With regard to the dataset compiled by Walmsley, it was concluded that the Panofsky–Dutton formula (see Table I) performs better than others used in comparison. Note that only the downwind roughness length enters the Panofsky–Dutton formula. We also wish to emphasize Jackson's conclusion that the rate of boundary-layer growth does not depend on the type of the transition (rough-to-smooth or smooth-to-rough) but only on a scale that is a combination of two surface roughnesses. All those data correspond to neutral stability conditions.

A new wind-tunnel simulation of the roughness change under neutral conditions was reported recently (Cheng and Castro, 2002). Although the emphasis was on the near-wall roughness sublayer the IBL development and wind profile modification were studied as well.

4. A Revised Model

In Miyake (1965), the 'diffusion analogy' was applied to the atmosphere under neutral stability and used in studies of smoke plumes spreading within the turbulent boundary layer of the atmosphere. The smoke plume spread was thought of in conditions of a homogeneous lower boundary with scaling parameters pertaining to that boundary only. IBL development intrinsically involves interactions of dissimilar surfaces. In what way are the scales of the problem altered and how do they enter the picture? This question has to be addressed along with consideration of additional processes that appear solely due to the inhomogeneity of the problem domain.

Savelyev and Taylor (2001) revisited the derivation of the widely used Panofsky–Dutton formula and proposed some changes to it. Let us summarize briefly their arguments and results. Having the expression for the IBL rate of growth established in the form

$$U(\delta)\frac{d\delta}{dx} = A\sigma_w, \tag{10}$$

surface-layer scaling parameters are used in representations of mean speed and vertical velocity variance. If we use the upstream velocity scale, u_*, it will cancel on both sides of the equation. If the velocity variance is to be that of the modified flow, as was assumed by most of authors whose papers we reviewed, more assumptions are needed in the expressions that relate $\sigma_w(z)$ and $U(z)$ with the velocity scale of the downwind region. For example, Panofsky and Dutton (1984) hypothesized that in the region of modified flow

$$U(z) = \left(\frac{u_{*D}}{\kappa}\right)\ln\frac{z}{z_{0D}}$$

and

$$\sigma_w = 1.25u_{*D},$$

where u_{*D} is a velocity scale.

We took a different view and persisted with the basic 'diffusion analogy' propositions, assuming that the mixing is taking place at the top of the IBL where the velocity variance is that of the incoming flow and upwind region scales are used to calculate σ_w at $\delta = \delta(x)$ as well as $U(\delta)$. Note that no assumption was needed with respect to the shear stress behaviour inside the IBL. The resulting model equation

$$\left(\frac{u_{*U}}{\kappa}\right)\ln\left(\frac{\delta}{z_{0U}}\right)\frac{d\delta}{dx} = 1.25Au_{*U} \tag{11}$$

was a basis for obtaining values of A in each experiment from the dataset by means of regression analysis. Different experiments had different values of z_{0D}/z_{0U} and an empirical relationship

$$A = 1.0 + 0.1 \ln \frac{z_{0D}}{z_{0U}} \tag{12}$$

was established (Savelyev and Taylor, 2001).

4.1. Mean vertical velocity caused by local gradients. its role and approximation

Upon insertion of Equation (12) back into the original model (Equation (6)) and keeping in mind that $\sigma_w = 1.25 u_{*U}$ one can obtain

$$\frac{d\delta}{dt} = \sigma_w + 0.1 \ln\left(\frac{z_{0D}}{z_{0U}}\right)(1.25 u_{*U}). \tag{13}$$

It seems natural to look for a process other than diffusion to explain the second term on the right-hand side. Local horizontal gradients of the mean wind speed that arise after a step change in surface conditions cause a mean vertical velocity to appear due to continuity constraints. This was not taken into account in the 'diffusion analogy', probably because the IBL formation was compared to the smoke plume diffusion above a homogeneous infinite plane and there is no mean vertical displacement velocity in this case. We can augment Equation (6) by another factor that in our opinion influences the apparent IBL growth, namely mean vertical velocity $W(\delta)$, so that

$$\frac{d\delta}{dt} = A_1 \sigma_w + A_2^* W(\delta). \tag{14}$$

Further, in 2D situations, we can determine the vertical velocity from the horizontal velocity gradient through the continuity equation, i.e.,

$$W(\delta) = -\int_{z_{0D}}^{\delta} \frac{\partial U}{\partial x} dz, \tag{15}$$

where we assume $U = 0$ at $z = z_{0D}$. Using characteristic values in the above equation we can write

$$W(\delta) \sim -\frac{\Delta U}{L_x} L_z, \tag{16}$$

where L_z and L_x are vertical and horizontal scales taken equal to δ and x (fetch), respectively. As for ΔU, the characteristic difference between mean wind speed of incoming flow and the unbalanced flow in the IBL we offer the following estimate. Asymptotically, for large x, the equilibrium value of u_{*D} will be close to the upstream value u_{*U}. We assume that $\widehat{U}_D(z) = \kappa^{-1} u_{*U} \ln (z/z_{0D})$ is a reference value for the mean profile of the unbalanced flow in the IBL. Ignoring the region very close to the ground, we assume that the contributions to the integral of the Equation (15) arise mainly from the upper layers. We also note that

$$U_U - \widehat{U}_D = \left(\frac{u_{*U}}{\kappa}\right) \ln \frac{z}{z_{0U}} - \left(\frac{u_{*U}}{\kappa}\right) \ln \frac{z}{z_{0D}} = \left(\frac{u_{*U}}{\kappa}\right) \ln \frac{z_{0D}}{z_{0U}} \tag{17}$$

is independent of z for adiabatic flow. For diabatic flow this is not so and in general we will assume that ΔU is proportional to $U_U - \widehat{U}_D$ at $z = \delta$, and write

$$-\Delta U \sim (U_U - \widehat{U}_D)(\delta). \tag{18}$$

In neutral flow Equation (14) becomes

$$\left(\frac{u_{*U}}{\kappa}\right) \ln\left(\frac{\delta}{z_{0U}}\right) \frac{d\delta}{dx} = A_1 \sigma_w + A_2 \left(\frac{u_{*U}}{\kappa}\right) \frac{\delta}{x} M, \tag{19}$$

where $M = \ln(z_{0D}/z_{0U})$. Employing the relationship $\sigma_w = Cu_*$, one can obtain

$$\frac{d\delta}{dx} = \left(C\kappa A_1 + A_2 \frac{\delta}{x} M\right) \left(\ln \frac{\delta}{z_{0U}}\right)^{-1} \tag{20}$$

and solve it numerically, provided values of C, A_1 and A_2 are known. Or, we can rearrange it slightly differently as,

$$\frac{d\delta}{dt} = \left(A_1 + \left(\frac{A_2}{C\kappa}\right)\left(\frac{\delta}{x}\right) M\right) \sigma_w \tag{21}$$

to show that $A = A_1 + A_2 (C\kappa)^{-1} (\delta/x) M$, i.e. A is a function of $(\delta/x)M$. In Figure 4 we plot the set of best fit values of individual coefficients A mentioned above (see Savelyev and Taylor, 2001 for more details on how they were obtained) against $(\delta/x)M$. The value for δ/x was taken as the average for the particular experiment. The best fit line for the points is close to

$$A = 1.0 + (\delta/x)M, \tag{22}$$

which suggests we should take $A_1 = 1$ and $A_2 = C\kappa$ ($= 0.5$ if $C = 1.25$). In order for Equations (12) and (22) to be consistent the mean value of the δ/x for the experiments considered should be close to 0.1. Indeed this is the case. The final form of the expression for the IBL growth rate in adiabatic flow is then,

$$\frac{d\delta}{dx} = C\kappa \left(1 + \left(\frac{\delta}{x}\right) \ln \frac{z_{0D}}{z_{0U}}\right) \left(\ln \frac{\delta}{z_{0U}}\right)^{-1}. \tag{23}$$

Let us return to the scaling of Equation (6). We choose L_x and L_z to be x and $\delta(x)$, although $W(\delta)$ really depends on the whole column $(0 < z < \delta)$ rather than just the $[x, \delta(x)]$ point. The ratio δ/x is a mean slope of the IBL interface. The local value of this ratio at the considered point is $d\delta/dx$. With this scaling we have, instead of Equation (19), the following expression

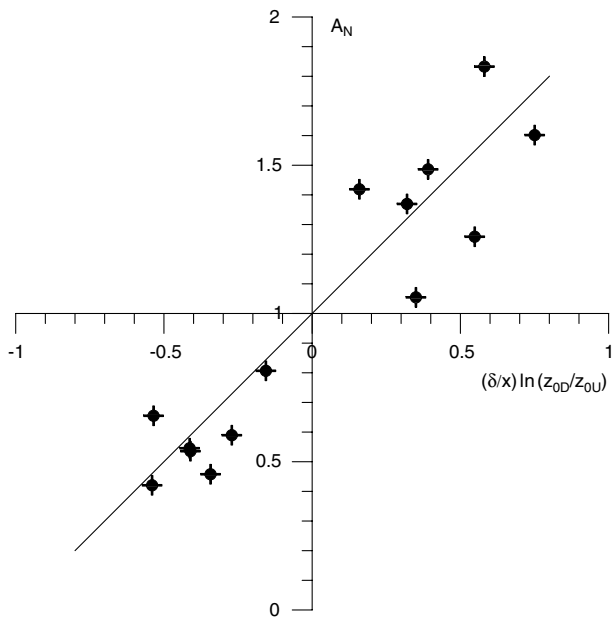

Figure 4. Proportionality coefficient A (in $d\delta/dt = A\sigma_w$) versus $(\delta/x)M$. Solid line represents $A = 1.0 + (\delta/x)\ln(z_{0D}/z_{0U})$.

$$\left(\frac{u_{*U}}{\kappa}\right)\ln\left(\frac{\delta}{z_{0U}}\right)\frac{d\delta}{dx} = A_1\sigma_w + A_2\left(\frac{u_{*U}}{\kappa}\right)M\frac{d\delta}{dx}. \tag{24}$$

Taking A_1, A_2 and σ_w as above and rearranging terms one obtains

$$\frac{d\delta}{dx} = C\kappa\left(\ln\frac{\delta}{z_{0U}} - C\kappa\ln\frac{z_{0D}}{z_{0U}}\right)^{-1}. \tag{25}$$

This equation can be integrated to produce an implicit formula for the neutral IBL depth, namely

$$\delta\left[\ln\frac{\delta}{z_{0U}} - \left(1 + C\kappa\ln\frac{z_{0D}}{z_{0U}}\right)\right] = C\kappa x + C^*, \tag{26}$$

where C^* is a constant of integration that we take equal to 0 (see discussion in Savelyev and Taylor, 2001). It is interesting to note that with $C = 1.25$ it follows that $C\kappa = 0.5$ and

$$\ln\frac{\delta}{z_{0U}} - 0.5\ln\frac{z_{0D}}{z_{0U}} = \ln\frac{\delta}{\sqrt{z_{0U}z_{0D}}}, \tag{27}$$

which means that Equation (26) predicts the same height regardless of the order of the adjacent surfaces:

$$\delta\left(\ln\frac{\delta}{\sqrt{z_{0U}z_{0D}}} - 1\right) = 0.5x. \tag{28}$$

Note that the length scale $\sqrt{z_{0U}z_{0D}}$ was also used by Deaves (1981).

Let us emphasize differences in the derivation of the Savelyev and Taylor (2001) expression and Equations (23) and (28). The first was obtained as a result of an empirical fit of experimental data into Miyake's model that relies on vertical turbulent velocity to transport roughness change impacts upwards from the surface. The next two formulae result from performing similar fitting procedures on the augmented model of the IBL growth. The mean vertical turbulent velocity due to local horizontal gradients was introduced into the model. Continuity constraints were employed to obtain a characteristic value for the mean vertical velocity through gradients of mean horizontal velocity. A choice of vertical and horizontal scales as IBL height and distance from the change in surface conditions led to the final expression (Equation (23)). The fact that the two scales were combined into the ratio allowed us to test an alternative scaling; namely the local value of the interface slope, $d\delta/dx$, was used in the model instead of δ/x.

The neutral IBL heights predicted by three formulae: Savelyev and Taylor (2001), Equations (23) and (28) for various steps in roughness are plotted in Figure 5. The local value of the interface slope $d\delta/dx$ as a multiplier to M results in slightly higher IBLs than the mean slope value δ/x in the case of a rough-to-smooth transition and vice versa in the smooth-to-rough case. The discrepancy increases with fetch though not significantly. The behaviour of the height predicted with a constant multiplier to M is similar for the longer fetches but it is reversed close to the roughness change line. The second distinction is a significant deviation of the $A = 1.0 + 0.1M$ line from two other formulae for large positive values of M.

4.2. Comparison of predicted and measured IBL heights

An independent experiment (in a sense that it is not a part of the dataset used to obtain empirical relationship for the coefficient A) was utilized as a test. In Echols and Wagner (1972) the average IBL heights measured at a tower installed 90 m from the Gulf of Mexico shore were reported as 7.2 m in the daytime and 5.9 m at night. IBL thickness was determined from the mean wind profiles as the intersection of two logarithmic parts. Air flow was from the sea ($z_{0U} = 3 \times 10^{-6}$ m) onto a sand beach ($z_{0D} = 3 \times 10^{-2}$ m). Daytime conditions were characterized as neutral or near neutral while nighttime profiles indicated stable stratification. We calculated the IBL extent at the distance of 90 m according to various short fetch formulae and results are presented in Table II. Raabe's (1983) equation produced the closest value but the heights in the table are more an illustration of the range of values (3.3–12.4 m) that different approaches can produce rather than a demonstration of the accuracy of a particular equation. More measurements are

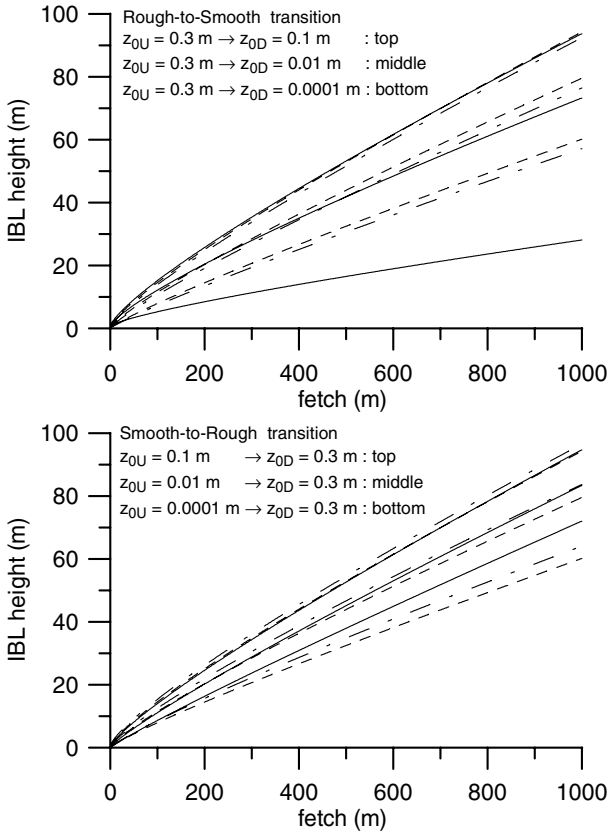

Figure 5. Prediction of the neutral IBL thickness according to three formulae: Savelyev and Taylor (2001) (solid line), Equation (23) (dash–dot line) and Equation (28) (dash line). Roughness change parameter $M[\equiv \ln(z_{0D}/z_{0U})] = \pm 1.1, \pm 3.4, \pm 8.0$.

required to draw statistically well-grounded conclusions. It is worth mentioning the slight hump between the sea and the tower that, in the opinion of Echols and Wagner (1972), may 'have some effect on the observed height', i.e. the recorded thickness could be larger than in a pure plane case due to this topographic effect.

Overall we found that including the effects of a streamline displacement provided a very satisfactory model of IBL growth in neutrally stratified flow.

5. The IBL Height in Diabatic Flow

We now consider the situation of a simple two-dimensional flow from one homogeneous surface to another with a sharp change in roughness and changes in sensible and latent heat flux. The upstream flow may be neutral or

TABLE II

The neutral IBL height at 90 m distance according to various short fetch formulae (δ). Measured height (daytime) (δ_m) was 7.2 m (Echols and Wagner, 1972). Roughness change is estimated to be from 3×10^{-6} to 3×10^{-2} m.

Author(s)	δ (m)	$\delta - \delta_m$ (m)	$\frac{\lvert\delta-\delta_m\rvert}{\delta_m}$ (%)
Jegede and Foken (1999)	3.3	−3.9	54
Townsend (1966)	4.0	−3.2	44
Wood (1982)	5.1	−2.1	29
Equation (26)	5.1	−2.1	29
Townsend (1965)	5.5	−1.7	24
Pendergrass and Aria (1984)	5.8	−1.4	19
Jackson (1976)	5.8	−1.4	19
Equation (23)	5.9	−1.3	18
Savelyev and Taylor (2001)	6.4	−0.8	11
Raabe (1983)	7.9	0.7	10
Radikevitsch (1971)	8.0	0.8	11
Elliott (1958)	8.6	1.4	19
Panofsky and Dutton (1984)	9.5	2.3	32
Panofsky (1973)	11.0	3.8	53
Miyake (1965)	12.4	5.2	72

diabatic. Once again we consider the imbalance caused by the new underlying surface as an entity diffused and displaced upwards within the incoming flow, so that

$$\frac{d\delta}{dt} = A_1 \sigma_{wU} + A_2^* W(\delta), \tag{29}$$

where A_1 and A_2^* are constants.

Values of the vertical velocity variance in the incoming flow are available through the flux–variance functions of Monin–Obukhov Similarity Theory, viz.

$$\frac{\sigma_w}{u_*} = 1.25\left(1 - 3\frac{z}{L}\right)^{1/3} \tag{30}$$

for the convective surface layer ($-2 \leq z/L \leq 0$) (Panofsky et al., 1977). Note that the multiplier on the right-hand side is 1.25 (1.3 in the original paper) which gives our assumed neutral stability limit equal to that value. For stable stratification σ_w/u_* is usually considered to be constant in the range of z/L from 0 to 0.8. We adopted the expression for $0 \leq z/L \leq 0.8$,

$$\frac{\sigma_w}{u_*} = 1.25. \tag{31}$$

We also need to know the mean wind speed at the IBL interface to proceed with calculations of the IBL growth rate. It is common practice to use stability correction functions, $\psi_m(\zeta)$, in the equation for the wind speed in the diabatic surface layer, i.e.,

$$U(z) = \frac{u_*}{\kappa}\left[\ln\left(\frac{z}{z_0}\right) - \psi_m(\zeta) + \psi_m(\zeta_0)\right], \tag{32}$$

where $\psi_m(\zeta) = \int_{\zeta_0}^{\zeta}[1 - \phi_m(\zeta)](d\zeta/\zeta)$. Here $\phi_m(\zeta)$ is a flux-profile similarity function, with $\zeta = z/L$ and $\zeta_0 = z_0/L$. Under the assumption of small surface roughness and consequently, small ζ_0, $\psi_m(\zeta_0)$ is usually dropped from the above equation. For an unstable surface layer Paulson (1970) obtained an expression

$$\psi_m(\zeta) - \psi_m(\zeta_0) = 2\ln\left(\frac{1+x}{2}\right) - \ln\left(\frac{1+x^2}{2}\right) - 2\tan^{-1}(x) + \frac{\pi}{2}, \tag{33}$$

where $x = (1-\gamma)^{1/4}$ and γ is a constant. For a stable surface layer

$$\psi_m(\zeta) - \psi_m(\zeta_0) \simeq \psi_m(\zeta) = -\alpha\zeta \tag{34}$$

(α is a constant) is widely accepted. Constants α and γ stem from the flux–profile functions. In the so-called Businger–Dyer model $\alpha = 5$ and $\gamma = 16$ (see e.g. Kaimal and Finnigan, 1994).

Now for diabatic flow we follow the same path as for the neutrally stratified atmosphere but here we must use the approximation to the vertical velocity by means of the difference in horizontal velocity at the interface $W(\delta) \sim (\delta/x)(U_U - \widehat{U}_D)$ calculated with corrections (33) or (34) implied by stability situations before and after the step change. Then

$$\begin{aligned} U_U - \widehat{U}_D &= \frac{u_{*U}}{\kappa}\left[\ln\frac{\delta}{z_{0U}} - \psi_m\left(\frac{\delta}{L_U}\right)\right] - \frac{u_{*U}}{\kappa}\left[\ln\frac{\delta}{z_{0D}} - \psi_m\left(\frac{\delta}{L_D}\right)\right] \\ &= \frac{u_{*U}}{\kappa}\left[\ln\frac{z_{0D}}{z_{0U}} - \psi_m\left(\frac{\delta}{L_U}\right) + \psi_m\left(\frac{\delta}{L_D}\right)\right]. \end{aligned} \tag{35}$$

Here, L_U and L_D are upwind and downwind values of the Obukhov length, respectively. At this point we are ready to write the general expression for IBL growth rate that will include neutral stratification as a special case, namely

$$\begin{aligned} &\left[\ln\frac{\delta}{z_{0U}} - \psi_m\left(\frac{\delta}{L_U}\right)\right]\frac{d\delta}{dx} = \\ &\quad A_1\kappa\frac{\sigma_{wU}}{u_{*U}} + A_2\frac{\delta}{x}\left[\ln\frac{z_{0D}}{z_{0U}} - \psi_m\left(\frac{\delta}{L_U}\right) + \psi_m\left(\frac{\delta}{L_D}\right)\right], \end{aligned} \tag{36}$$

where σ_{wU}/u_{*U} will follow from Equation (30) or (31) depending on upwind stability. Note that proportionality coefficients A_1 and A_2 could, in principle, be functions of stability as well but we have here assumed that they are the

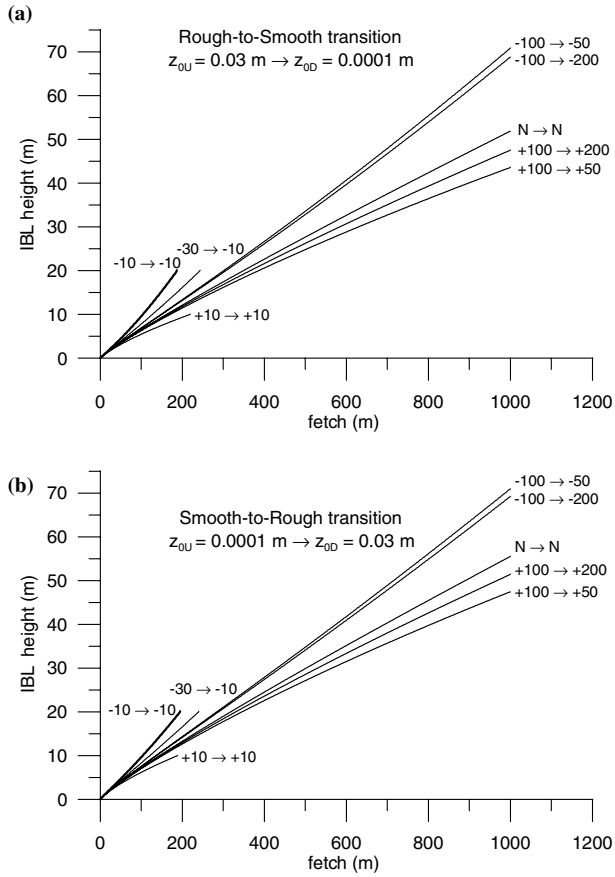

Figure 6. Diabatic IBL height in cases of similar stability situations upwind and downwind: (a) rough-to-smooth transition, (b) smooth-to-rough transition. Numbers near the curves indicate the change of the Obukhov length (L) value. N indicates neutral stratification.

same as for the neutral case. The Obukhov length, L_D, could well be a function of x, if, for instance, the step change was characterized by a step change in temperature rather than of heat flux. In this case, however, we would need to specify the forms of temperature and velocity profile within the IBL, in order to obtain the set of implicit equations necessary to solve for $\delta(x)$.

The IBL heights for different values of upstream and downstream Obukhov length that characterize stability effects are plotted in Figures 6a and b for roughness changes from 0.03 m to 0.0001 m or vice versa. Curves are drawn up to the height that limits the validity of flux–variance and stability correction functions; $\delta/L \leq 1$ if $L > 0$ and $\delta/|L| \leq 2$ if $L < 0$. Note that in Figures 6a and b there is no change in the type of stability situation. It is either a stable-to-stable or a convective-to-convective transition. The more unstable the stratification of the incoming flow, the faster the IBL grows. If

the upwind surface layer is stable the IBL upward spread is reduced with increasing stability. The closer the downwind stratification is to neutral the smaller is its influence on the IBL growth rate. Figure 6a reflects a rough-to-smooth transition while Figure 6b depicts the IBL growth in smooth-to-rough flow. The upwind roughness length z_{0U} and Obukhov length L_U appear more than once in Equation (36). Their action depends on the distance from the step change. Close to the discontinuity, the IBL grows faster if the flow is onto a rougher surface as in the case of neutral stratification. Further on, the influence of the roughness change is significantly less than the stability effects. It can be seen that rough-to-smooth and smooth-to-rough cases look similar. Changes in Obukhov length dominate over the change in roughness.

Examples of IBL height in situations when the stability parameter changes sign are plotted on Figure 7. Unstable thermal stratification upwind causes the interface to be generally higher than in the case of neutral-to-neutral transition and the opposite is true for the stable upwind stratification. One can see that the line for neutral–neutral transition (shown as a dashed line) is not a line of symmetry in the sense that IBL growth is enhanced more when stability changes from unstable to stable than it is damped in the opposite situation. Fairly stable conditions downwind ($L \leq 10$ m) result in IBL heights being lower than in adiabatic flow even when the upwind stratification is unstable. The line of $L = -50$ m to $L = 10$ m is stretched above the limit of validity to better show the tendency of the IBL growth.

In obtaining these results solution of Equation (36) is achieved by means of the 4th-order Runge–Kutta algorithm starting from some value of the IBL height at the point close to the discontinuity. The term with $\ln(z_{0D}/z_{0U})$ on the right-hand side may render $d\delta/dx$ to be negative close to $x = 0$ in the case of rough-to-smooth transition. It is obvious that the integration should start from the point where $d\delta/dx > 0$. To find this point we first deduce from the neutral stability case (Equation(25) with $C = 1.25$ and $\kappa = 0.4$) (since close to $x = 0$ the value of δ/L is also close to 0 too, i.e., close to neutral), that

$$\frac{d\delta}{dx} = 0.5\left(1 + \frac{\delta}{x}\ln\frac{z_{0D}}{z_{0U}}\right)\left(\ln\frac{\delta}{z_{0D}}\right)^{-1}. \tag{37}$$

Thus for $d\delta/dx > 0$ we need $(\delta/x)|\ln(z_{0D}/z_{0U})|$ less than 1, or $\delta < x/|\ln(z_{0D}/z_{0U})|$. We then insert this inequality into the expression for the neutral IBL height (Equation(28)),

$$\delta\left(\ln\frac{\delta}{\sqrt{z_{0U}z_{0D}}} - 1\right) = 0.5x \tag{38}$$

to obtain

$$\frac{x}{|\ln(z_{0D}/z_{0U})|}\left(\ln\frac{x}{\sqrt{z_{0U}z_{0D}}\,|\ln(z_{0D}/z_{0U})|} - 1\right) > 0.5x \tag{39}$$

or

$$\frac{x}{\sqrt{z_{0U}z_{0D}}\mid \ln(z_{0D}/z_{0U})\mid} > \exp(1 + 0.5 \mid \ln(z_{0D}/z_{0U})\mid).$$

Taking into account that $\mid \ln(z_{0D}/z_{0U})\mid = \ln(z_{0U}/z_{0D})$ in the rough-to-smooth transition case it follows that one should have

$$x > \exp\,(1)\sqrt{\frac{z_{0U}}{z_{0D}}}\sqrt{z_{0U}z_{0D}}\ln\frac{z_{0U}}{z_{0D}} = \exp\,(1)z_{0U}\ln\frac{z_{0U}}{z_{0D}}.$$

In order to have the same starting point for both smooth-to-rough and rough-to-smooth transitions the last inequality can be generalized to

$$x > \max(z_{0U}, z_{0D})\exp\,(1)|\ln(z_{0D}/z_{0U})|. \tag{40}$$

Numerical checks showed that taking

$$x \geq 1.01\ \exp\,(1)|\ln(z_{0D}/z_{0U})|\max(z_{0U}, z_{0D}) \tag{41}$$

ensures that $d\delta/dx > 0$ for a wide range of roughness and stability changes. The IBL height at this starting point is calculated from Equation (28), the neutral case.

The scenario that we have considered, with flow from one infinite plane to another unbounded one, is hypothetical. In order to apply our approach in practical situations one would need, in addition to specifying surface roughnesses, to suggest an approximation to the Obukhov lengths that we used to characterize thermal stratification. In the case of only one surface discontinuity and a sufficiently long extent of the first surface (with the steady state requirement also in effect) the determination of the Obukhov length L_U in the incoming

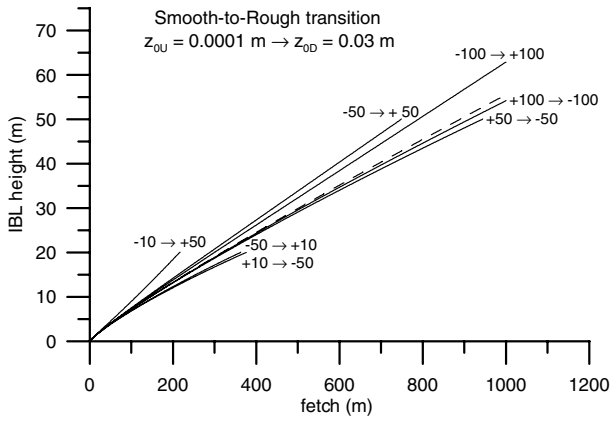

Figure 7. Diabatic IBL height in cases of opposite upwind and downwind stabilities, smooth-to-rough transition. Numbers near the curves indicate the change of the Obukhov length (L) value. Dashed line is for neutral stability both upwind and downwind.

flow is well defined and can be determined. What is not clear is how we can provide the value of L_D. A constant value for this surface-layer scale implies a constant stress layer in equilibrium with the underlying surface whereas what we have is a flow in a phase of adjustment to a new surface. For the momentum flux, the major adjustment is confined to the first several metres of fetch. An internal equilibrium layer (IEL) was reported by many investigators. These facts allows us to speculate that with increasing distance from a discontinuity the apparent value of the Obukhov length (constructed from values of fluxes measured at this distance and within the IEL) will approach an equilibrium. The requirement for the measurement height z_m to be smaller than the IEL height (δ_e) can provide us with a criterion for the distance where measurements should be taken. Assuming that δ_e is proportional to the IBL height, i.e.,

$$\delta_e = C\delta, \tag{42}$$

with C being some constant, we can use Equation (28) to deduce that

$$x \geq 2C^{-1}z_m\left(\ln\frac{C^{-1}z_m}{\sqrt{z_{0U}z_{0D}}} - 1\right) \tag{43}$$

should hold in a neutral surface layer in order that $z_m \leq \delta_e$. As a starting point, $C = 0.1$ can be recommended for neutral stratification. In diabatic flow the IBL height differs from its neutral counterpart but we can retain the neutral value in Equation (42) and adjust the coefficient C accordingly. Thus we will be able to use the same criterion (43) but with a proper coefficient $C(\zeta)$ that is to be determined from experiments.

There are quite a few measurements of the diabatic IBL that we can use to verify our assumptions. One of the experiments is described in Ogawa and Ohara (1985). Along with IBL ($\delta = 12\,\mathrm{m}$) and IEL ($\delta_e = 6.5\,\mathrm{m}$) heights at fetch = 160 m the fluxes of momentum and sensible heat were measured. We chose this fetch on the grounds that the sensors were situated inside the equilibrium layer and as such the calculated Obukhov length ($L_D \simeq -9\,\mathrm{m}$) is close to the equilibrium value. The case under study was 'near-neutral flow aloft penetrating inland with a strongly unstable layer forming below'. It was noted also that near-neutral flow above the sea was on the slightly unstable side. We can estimate L_U employing an empirical formula for the calculation of the sensible heat flux (H, in W m^{-2}) above the sea as,

$$H = 14.654(T_w - T_a), \quad \textit{if} \quad T_w > T_a$$

(Shuleikin, 1953), where $T_w - T_a$ is the sea–air temperature difference in kelvin. With $T_w - T_a \simeq 0.3\,K$ we obtained $L_U \simeq -140\,\mathrm{m}$. The IBL heights produced by our model were: 10.2 m (neutral to $L_D = -9\,\mathrm{m}$) and 10.5 m ($L_U = -140\,\mathrm{m}$ to $L_D = -9\,\mathrm{m}$). That was within 15 and 12.5 % of the measured values, respectively.

6. Conclusions

From a review of many alternative theories and formulations of the internal boundary-layer depth, and focussing on the neutrally stratified boundary layer, we conclude that the Miyake diffusion analogy is a good foundation for IBL depth prediction. It can, however, be improved by the addition of a displacement effect associated with flow convergence or divergence related to the roughness-change-induced deceleration or acceleration.

Estimates of the displacement term in terms of $\Delta U d\delta/dx$ where $\Delta U \propto \ln(z_{0D}/z_{0U})$ lead to modified IBL formulae that perform well in comparison with previously published expressions. The method can be extended to diabatic (non-neutral stability) conditions.

Modifications of a diabatic air flow due to a change in surface conditions can be predicted based on these IBL height formulae provided some assumptions are made on the form of mean flow properties profiles. Part II will be devoted to that subject.

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